

In the previous lecture, we showed that if an inverse of a matrix exists, then it is unique. We have also showed that a product of invertible matrices is again invertible, that is, if A and B are two invertible matrices then AB is invertible, moreover $(AB)^{-1} = B^{-1}A^{-1}$. Note that given the latter, one can use induction to show that if $A_1, A_2 \dots A_n$ are invertible matrices, the product $A_1 A_2 \dots A_n$ is invertible with $(A_1 A_2 \dots A_n)^{-1} = A_n^{-1} \dots A_2^{-1} A_1^{-1}$.

Elementary matrices and row equivalence

Let A be an $m \times n$ matrix and e denotes one of the following operations (below R_i denotes the i -th row):

- $R_i \leftrightarrow R_j$ (interchanging rows)
- $R_i \rightarrow \lambda R_i$, where $\lambda \neq 0$ (multiplying a row by a non-zero scalar)
- $R_i \rightarrow R_i + \lambda R_j$, where $i \neq j$ (adding a multiple of the j -th row to the i -th row)

Then by $e(A)$ we mean the matrix obtained by applying operation e to matrix A .

Examples Let $A = \begin{pmatrix} 0 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$, if e is the operation $R_2 \rightarrow R_2 + 5R_1$, then $e(A) = \begin{pmatrix} 0 & 3 & 1 \\ 1 & 17 & 8 \end{pmatrix}$ and if e' is the operation $R_1 \leftrightarrow R_2$, then $e'(A) = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \end{pmatrix}$

Definition 22. Let I_m be the $m \times m$ identity matrix and e be an operation as above, then by an elementary matrix associated to e we mean a matrix obtained by applying e to the identity matrix I_m . That is, elementary matrices are matrix of form $e(I_m)$ where e is one of the operations above.

Examples Below are four elementary matrices of different sizes obtained by $R_2 \rightarrow 3R_2$, $R_1 \leftrightarrow R_2$, $R_2 \rightarrow R_2 + 2023R_4$, and $R_1 \rightarrow 1 \cdot R_1$, respectively.

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2023 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Proposition 9. Let A be an $m \times n$ matrix and e be one of the above operations. Then there exists an elementary matrix E such that $e(A) = EA$.

Proof. Take $E = e(I)$, and check for each of the operations that this is the case (as illustrated for operation $R_i \leftrightarrow R_j$ in class, the example below may also be of help). $\square$

Below, observe carefully what is happening with matrix multiplication EA .

Example On the very first example, we had that if $A = \begin{pmatrix} 0 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$, and e is the operation $R_2 \rightarrow R_2 + 5R_1$, then $e(A) = \begin{pmatrix} 0 & 3 & 1 \\ 1 & 17 & 8 \end{pmatrix}$. The elementary matrix E associated to $R_2 \rightarrow R_2 + 5R_1$ is $E = e(I_2) = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$ and the matrix multiplication $EA = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 0 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 1 \\ 1 & 17 & 8 \end{pmatrix} = e(A)$

Proposition 10. Each elementary matrix is invertible and its inverse is an elementary matrix of the same type.

Proof. The above operations are reversible, and so for each operation e there exists an operation f of the same type such that $e(f(A)) = A$ and $f(e(A)) = A$ for every matrix $m \times n$ A . Let E and F be elementary matrices corresponding to e and f , respectively, then we have that $(EF)A = A$ and $(FE)A = A$ for every matrix A . Since this is true for every matrix A , we must have that $EF = I_m$ and $FE = I_m$. $\square$

Definition 23. Let A and B be $m \times n$ matrices, then A is said to be row equivalent to B if there exists operations $e_1, e_2, \dots, e_k$ such that $A = e_k \dots e_1(B)$

Note that every matrix A is row equivalent to itself. Row equivalence is also symmetric, if A is row equivalent to B then we have that B is row equivalent to A , for if there exists $e_1, e_2, \dots, e_k$ such that $A = e_k \dots e_2 e_1(B)$ we have $B = e_1^{-1} e_2^{-1} \dots e_k^{-1}(A)$. If A is row equivalent to B and B row equivalent to C , then we also have that A is row equivalent to C . Indeed, this means there exists $e_1, e_2, \dots, e_k$ such that $A = e_k \dots e_2 e_1(B)$ and there exists $e'_1, e'_2, \dots, e'_l$ such that $B = e'_l \dots e'_2 e'_1(C)$ and so $A = e_k \dots e_2 e_1(e'_l \dots e'_2 e'_1(C))$ thus A is row equivalent to C .

Proposition 11. Every $m \times n$ matrix A is row equivalent to an $m \times n$ matrix in row echelon form.

Proof. Given matrix A , we can row reduce it to row echelon form using at each step the above operations and the Gaussian elimination algorithm, thus obtain our row equivalence. $\square$